MSc DATA SCIENCE

WhatsApp NLP & Conversation Analytics Dashboard

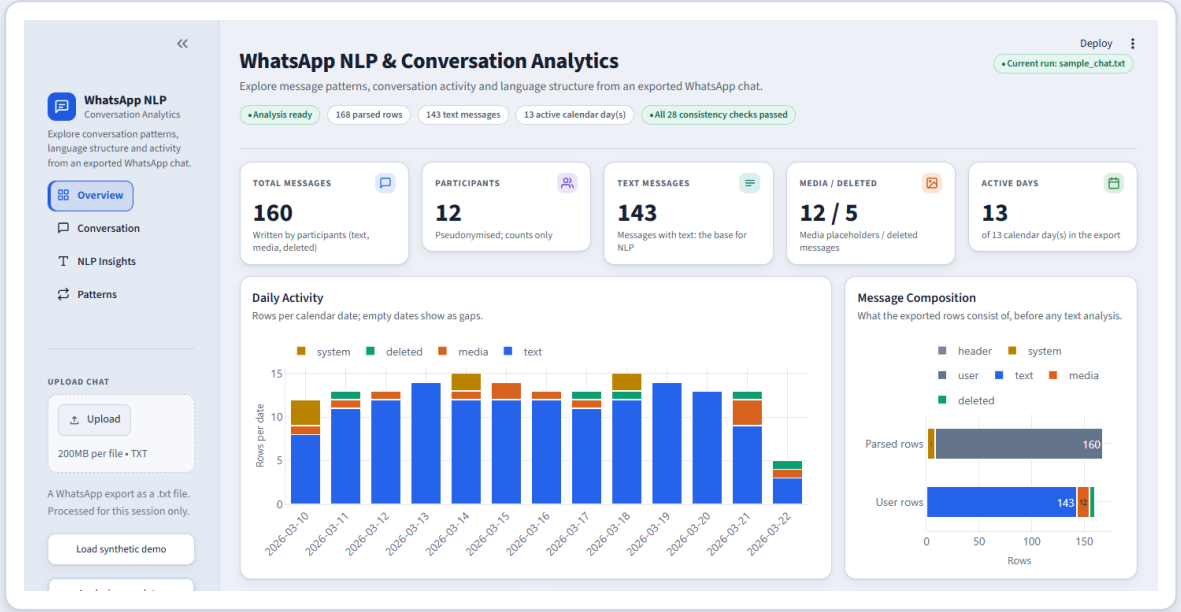
A privacy-aware pipeline and interactive dashboard for describing the language and activity structure of exported WhatsApp chats

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Overview page, fictional sample chat

Problem & Objectives

Why a raw chat export needs a careful, explainable analysis pipeline

PROBLEM

An exported WhatsApp chat is a raw text file. It is hard to see directly:

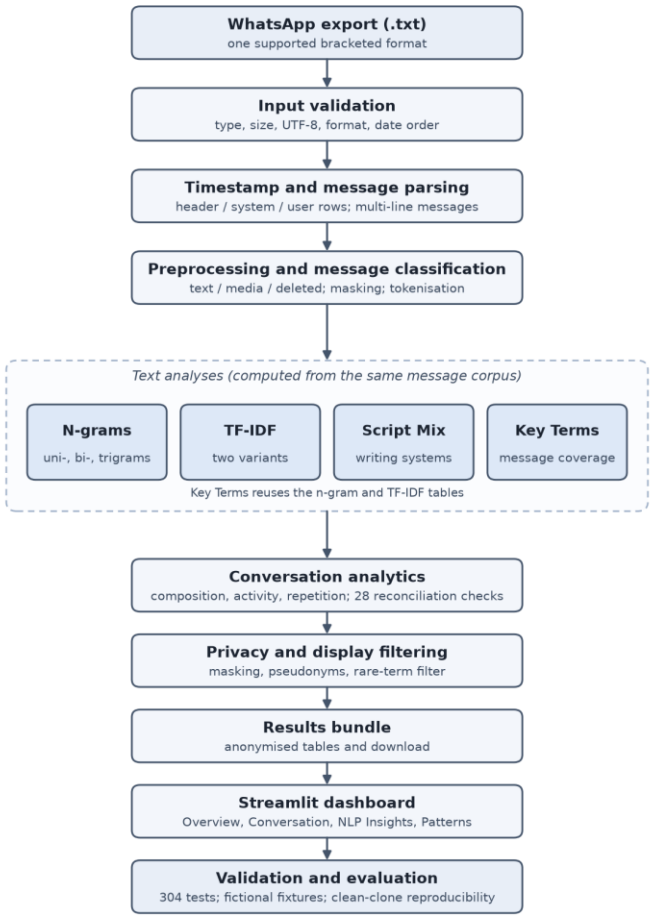
- Message patterns
- Frequently used terms
- Activity over time
- Repeated content
- Writing-system usage
- Conversation composition

OBJECTIVES

1. Build a robust WhatsApp parser and preprocessing pipeline
2. Apply NLP techniques: n-grams and TF-IDF
3. Analyse Script Mix and key conversational terms
4. Extract activity and repetition patterns
5. Present results in a privacy-conscious interactive dashboard

Data & Methodology

One supported export format, one reproducible pipeline



WHAT HAPPENS TO AN UPLOADED CHAT

1. **Validate and parse**
   Bracketed .txt export; header, system and user rows; multi-line messages kept together

2. **Preprocess and classify**
   Text, media or deleted; links and phone numbers masked; Unicode-aware tokens

3. **Analyse**
   N-grams, TF-IDF, Script Mix, Key Terms and conversation analytics (28 reconciliation checks)

4. **Filter and present**
   Privacy filter, anonymised results bundle, Streamlit dashboard

Public demonstration uses a fictional WhatsApp-style dataset for privacy and reproducibility.

NLP Analysis

N-GRAMS

- Unigrams
- Bigrams
- Trigrams
- All messages vs unique texts

TF-IDF

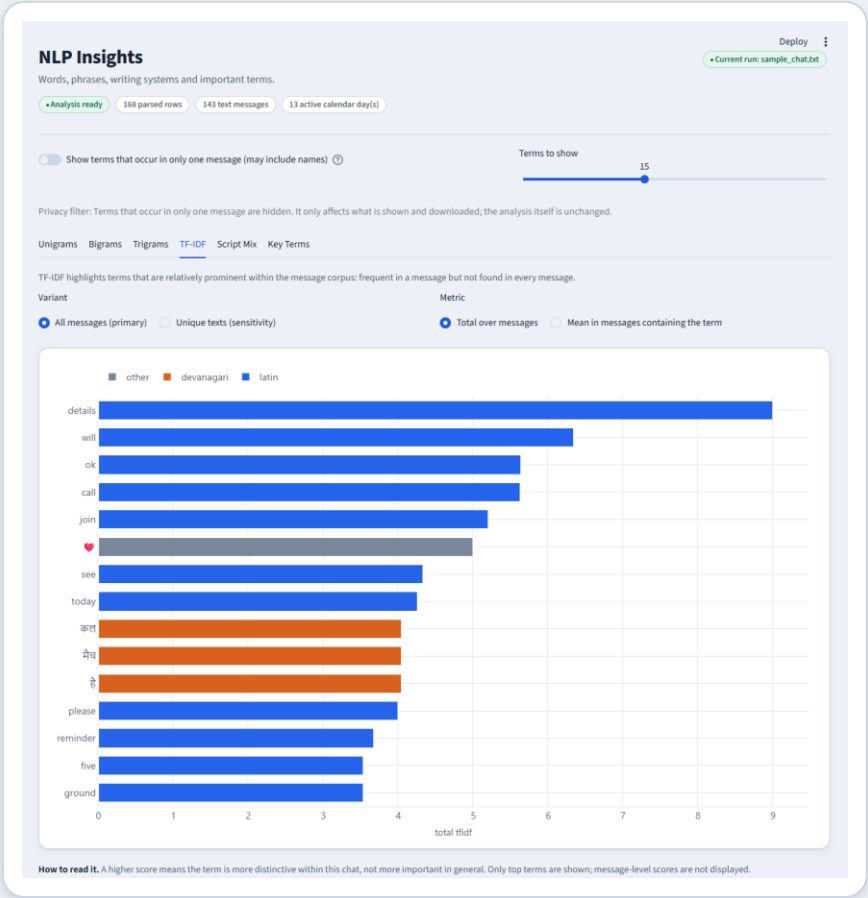
- Term distinctiveness
- Primary + unique-text variants
- Validated against scikit-learn

SCRIPT MIX

- Latin
- Devanagari
- Mixed-script
- No-alphabetic

KEY TERMS

- Message coverage
- Total occurrences
- Repeated-message effect



TF-IDF tab, fictional sample chat

Script Mix is a writing-system analysis, not language identification.

Conversation Analytics

Fictional sample dataset: 12 invented participants

TOTAL ROWS
168
parsed

USER ROWS
160
of 168 rows

TEXT MESSAGES
143
analysed

MEDIA ROWS
12
placeholders

DELETED ROWS
5
placeholders

PARTICIPANTS
12
pseudonymised

ACTIVE DAYS
13
calendar days

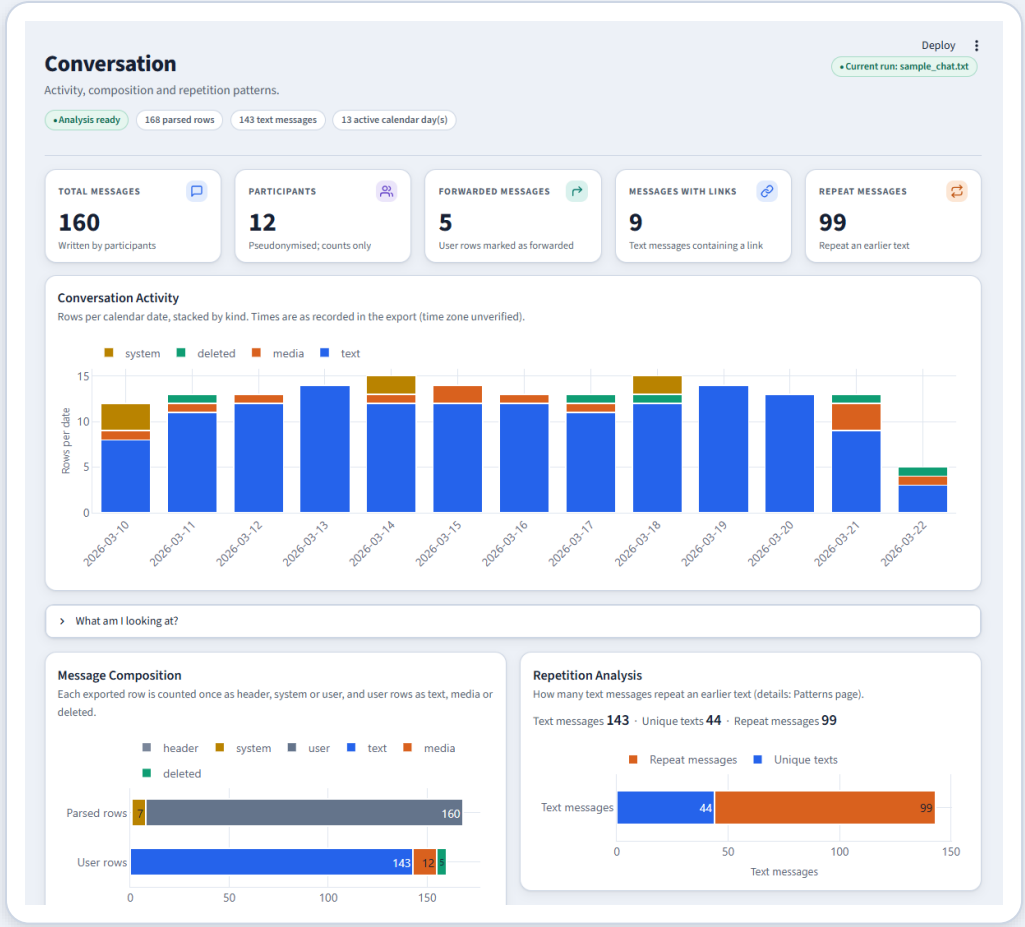
PEAK ACTIVITY
15 rows
on 2026-03-14

REPEATED TEXTS
99 / 143
text messages

UNIQUE TEXTS
44
after masking

TEMPLATES
35
repeated

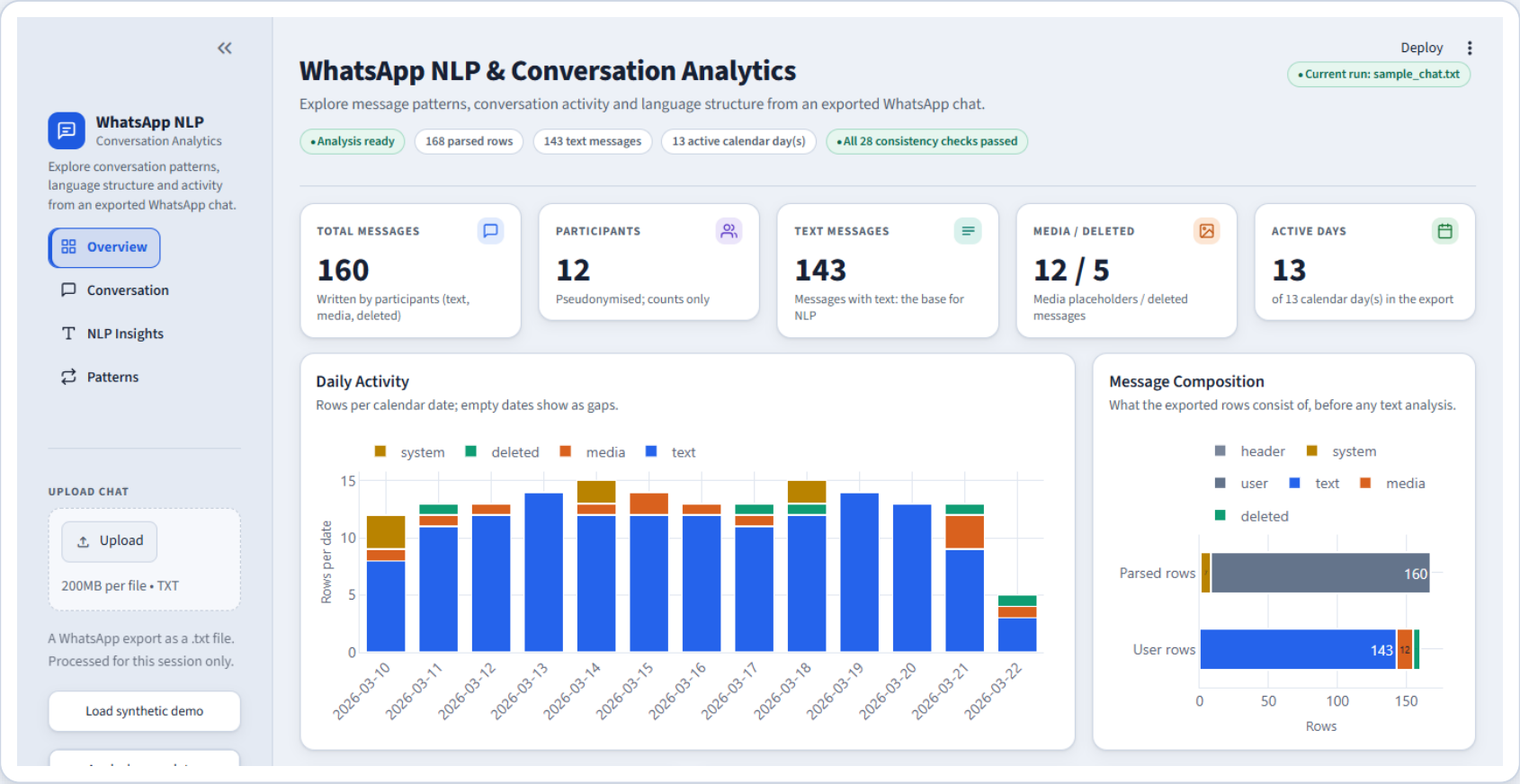
Fictional sample dataset



Conversation page, fictional sample chat

Interactive Dashboard

Streamlit + Plotly academic prototype; upload-driven, session-based analysis



Overview page loaded with the fictional sample chat (12 invented participants)

Privacy-conscious display | **Anonymised download** | **Optional theme image**

Overview

key figures, daily activity, composition, key findings

Conversation

activity, composition, repetition, hour and weekday tables

NLP Insights

six tabs: n-grams, TF-IDF, Script Mix, Key Terms

Patterns

repetition summary and anonymous templates

Validation & Limitations

What was checked, and what the results do not show

VALIDATION

- 304 tests passed in a clean clone
- 308 tests locally (4 are local-only)
- 28 reconciliation checks
- TF-IDF vs scikit-learn: max difference about 1.1×10^{-16}
- Committed outputs reproduced

PRIVACY

- Fictional public dataset
- Private chat excluded from the repository
- Privacy and display filtering
- Repository privacy scan
- No private chat results shown publicly

LIMITATIONS

- Supported export format is limited
- Android / no-bracket formats unsupported
- Script Mix is not language identification
- Privacy filter is not complete PII detection
- Public results are not population-level findings

Conclusion & Future Work

CONCLUSION

A reproducible pipeline for analysing WhatsApp-style exported conversations, with its limits stated.

- Preprocessing, n-grams and TF-IDF
- Script Mix and Key Terms
- Conversation analytics and an interactive dashboard
- Validation shows reproducibility and robustness
- A fictional public dataset enables safe demonstration

FUTURE WORK

1. Broader WhatsApp export-format support
2. Larger, real-world evaluation (group and iOS exports)
3. Richer multilingual analysis and stopword handling
4. Stronger personal-data detection
5. More robust large-scale TF-IDF evaluation

Thank you **Questions?**